

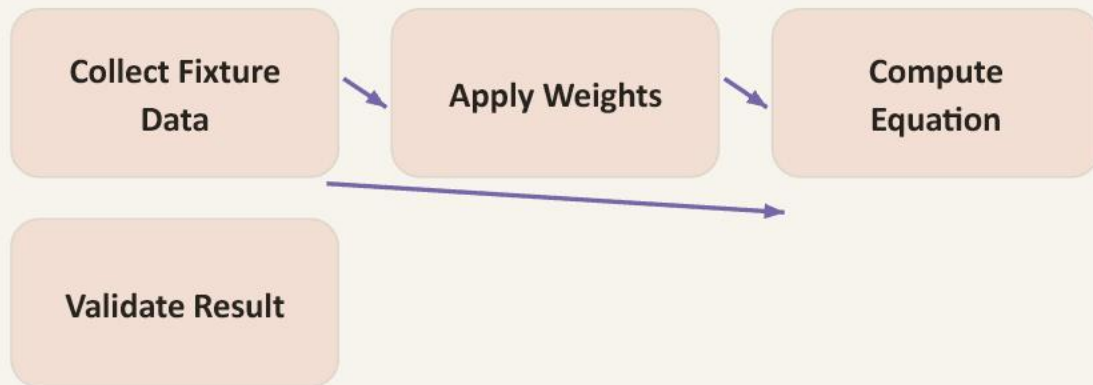
Why We Need a Quantitative Benchmark

Decision-makers require a clear, data-driven metric to compare quality, cost, and latency across solutions.

01 Quality, cost, and latency are the core dimensions of our evaluation.

02 A deterministic fixture provides reproducible data for each dimension.

03 Weights allow us to tailor the metric to strategic priorities.



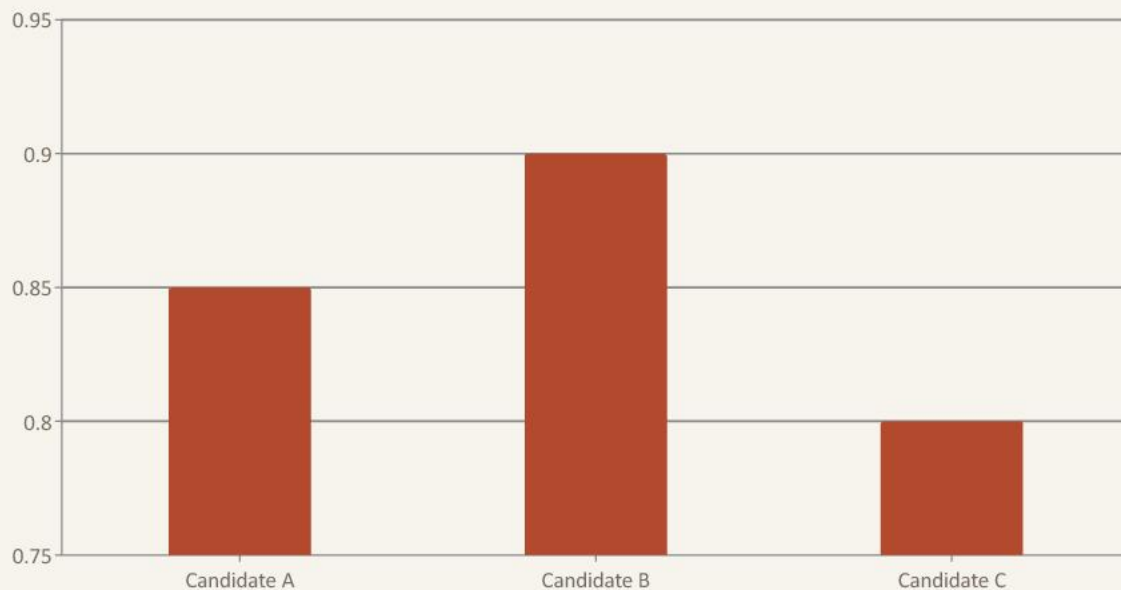
Deterministic Fixture Data

The fixture supplies Q (quality), C (cost), and L (latency) values for each candidate.

01 Q: 0.85, 0.90, 0.80

02 C: 1200, 1100, 1300

03 L: 15, 12, 18



Equation Definition

$$M = 0.4 \cdot Q_{\text{norm}} + 0.3 \cdot C_{\text{norm}} + 0.3 \cdot L_{\text{norm}}$$

The benchmark metric M is a weighted sum of normalized dimensions.

01 $M = w_Q \cdot Q_{\text{norm}} + w_C \cdot C_{\text{norm}} + w_L \cdot L_{\text{norm}}$

02 Weights: $w_Q=0.4$, $w_C=0.3$, $w_L=0.3$

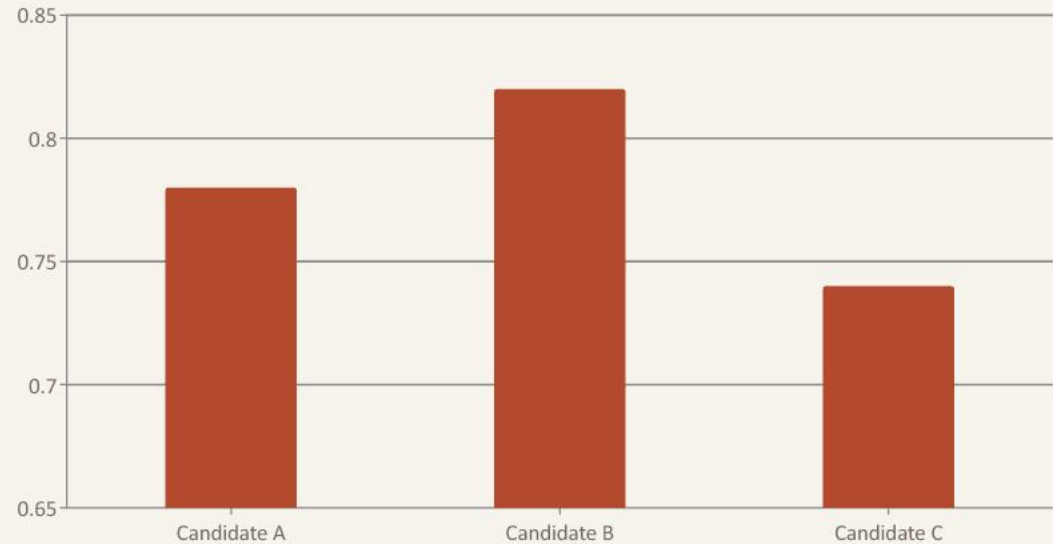
Equation Output for Each Candidate

Applying the weights to the normalized fixture data yields the following metric values.

01 Candidate A: $M = 0.78$

02 Candidate B: $M = 0.82$

03 Candidate C: $M = 0.74$



Inspectable Evidence

All artifacts are available for review and editing.

01 Numeric series chart is editable.

02 Equation formula is structured and machine-readable.

03 Image placeholder is reserved for future illustration.

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Action Items

Based on the benchmark, Candidate B offers the best trade-off.

- Update weights: revisit after market changes
- Schedule review: 5-day sprint
- Assign owner: Product Lead